

COLBY CADDEN

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Education

University of Calgary

September 2025 – Present

B.Sc. Mechanical Engineering / B.Comm. Finance; GPA: 4.0/4.0

Calgary, Alberta

- **Schulich Leader Scholarship** — \$120,000 national STEM scholarship recognizing academic excellence, leadership, and entrepreneurship.
- **President's Admission Scholarship** — awarded for an admission average of 95% or higher.

Salmon Arm Secondary

Graduated 2025

Dual Dogwood Diploma

Salmon Arm, British Columbia

- BC District/Authority Award — BC Achievement Scholarship — Governor General's Academic Medal

Engineering Projects

Sensor-Activated Robotic Arm | *Fusion 360, VS Code, ESP32, Arduino Nano, C++*

2025 – 2026

- Designed and iterated a sensor-activated desktop robotic arm using ESP32 and Arduino Nano microcontrollers, ToF sensors, servo-driven joints, and 3D-printed PLA components.

Remote-Controlled Rescue Vessel | *Raspberry Pi Pico, Fusion 360, Thonny, 3D Printing*

2026

- Led the mechanical design and fabrication of a remote-controlled rescue vessel built to navigate obstacles while transporting a payload.

G-Hacks 2026 Hackathon — First Place | *Python, GNSS Data Processing, High-Precision Positioning*

2026

- Processed raw real-time kinematic (RTK) and precise point positioning (PPP) satellite data in Python, applying geodetic and Cartesian coordinate transformations to produce the competition's most accurate positioning solution within 24 hours.

Cursor AI Hackathon | *Next.js, React, TypeScript, Tailwind, Gemini, Groq*

2026

- Built PrepDeck, an AI meal-planning platform with ingredient inventory, barcode scanning, recipe generation, structured meal plans, and reusable shopping lists through server-side API routes.

Leadership and Involvement

Engineering Team Member, Mechanical Lead, *EQOL*

2026 – Present

- Development of assistive technologies designed to improve daily tasks for people with disabilities, contributing CAD design, project pitching, prototyping, and electronics integration.

Student Representative, *University of Calgary*

2025 – Present

- Represented 900+ first-year engineering students by communicating feedback and collaborating with faculty to identify and implement solutions.

Athletic and Community Leadership, *Salmon Arm Secondary*

2021 – 2025

- Captained volleyball and ultimate frisbee teams, co-captained basketball, and competed on BC XC skiing talent squad.

Work Experience

Production Labourer, *Canoe Forest Products*

June 2026 – Present

- Operate large production machinery and troubleshoot equipment breakdowns to minimize downtime while maintaining safety and production targets.

Cone Picker, *Skimikin Seed Orchard*

2024 and 2025

- Harvested pinecones using ladders and hand tools while meeting production targets and maintaining safe work practices.

General Labourer, *Mt. Ida Tree Nurseries Ltd. and Sorrento Nurseries Ltd.*

2023 – 2025

- Packaged and handled nursery stock in fast-paced outdoor production environments.

Technical Skills and Certifications

Programming: Python, C++, MATLAB

Engineering Tools: Fusion 360, SolidWorks, Onshape, Cursor, VS Code, Microsoft Excel

Hardware and Fabrication: ESP32, Arduino Nano, Raspberry Pi Pico, Woodworking, Metalworking, Drafting

Certifications: DELF B2 French Language Proficiency; Economic Fundamentals for Leadership